BEFORE YOU READ

### The function families on one page

Every family on this page, grouped the way the page treats them. Polynomials and radicals open into their members.

- L0 **Function families**
  - L1 **Polynomial**
    - L2 **Constant** horizontal line, range {c} — $f(x) = c$
    - L2 **Linear** constant slope — $mx + b$
    - L2 **Quadratic** parabola with a vertex — $ax^2 + bx + c$
    - L2 **Cubic** S-curve, up to 2 turning points — $ax^3 + \ldots$
    - L2 **Higher degree** up to n zeros, n − 1 turns — degree n
  - L1 **Rational** asymptotes; holes possible — $P(x)/Q(x)$
  - L1 **Radical**
    - L2 **Square root** starts at the origin — $\sqrt{x}$
    - L2 **Cube root** S-curve through the origin — $\sqrt[3]{x}$
    - L2 **nth root, even n** domain [0, ∞)
    - L2 **nth root, odd n** domain $\mathbb{R}$
  - L1 **Absolute value** V-shape, vertex at the origin — $|x|$
  - L1 **Step** staircase, range $\mathbb{Z}$ — $\lfloor x \rfloor$, $\lceil x \rceil$
  - L1 **Exponential** growth or decay, asymptote y = 0 — $a \cdot b^x$
  - L1 **Logarithmic** inverse of exponential, asymptote x = 0 — $\log_a x$
  - L1 **Trigonometric**
    - L2 **Sine and cosine** period 2π, range [−1, 1]
    - L2 **Tangent** period π, vertical asymptotes

TAKEAWAY

### Recognise the shape, name the family

Each family leaves a characteristic shape on the graph: lines, parabolas, S-curves, asymptotes, half-curves, V-shapes, staircases and waves.